Introduction to Probability and Statistics

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Syllabus

- Probability
- Inferential Statistics
  - Estimation
  - Nonparametric statistics
  - Bayesian Statistics

AIMS African Institute for Mathematical Sciences RWANDA

Lecture 1: Probability I

Introduction to Probability and Statistics

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Random experiments – Events

- Probability provides a mathematical framework for the study of random phenomena or **random experiments**.
- The goal is to measure the chance of **outcomes** that are produced by a random experiment.
- The collection of all the possible outcomes is called the **sample space** or **universe**.

  It is usually denoted Ω.
- The **elementary outcomes** are denoted ω.
- An **event** is a subcollection of outcomes from Ω.

Example

- {*Rolling a die* } is a random experiment.
- The universe for this experiment is:

$$\Omega = \left\{ ⚀, ⚁, ⚂, ⚃, ⚄, ⚅ \right\}$$

  It has 6 elements.

- {*The face containing three dots appears*} is a possible elementary outcome.

$$\omega = \left\{ ⚂ \right\}.$$

- {*The result is odd* } is an example of event .

$$A = \left\{ ⚀, ⚂, ⚄ \right\}$$

  This event has 3 elements.

Events as sets

Exercise

- Random experiment: Toss a fair coin 10 times.

  Describe Ω the sample space.

- Let A be the event *'At least one head occurs'*.

  Give some examples of outcomes from A.

- Let T_j be the event that tails occurs on the j^{th} toss.

  Describe T_j.

Events as sets

Solution

- The sample space Ω contains elements of the form $(c_1, c_2, \ldots, c_{10})$ where $c_i = H$ or T, c_i being the result of the j^{th} toss.

  $$\Omega = \left\{(c_1, c_2, \ldots, c_{10}) \mid c_i = H \text{ or } T, i = 1, \ldots, 10\right\}.$$

  Example : $(T, T, H, T, H, H, T, T, T, T) \in \Omega$.
  Remark that the number of elements in Ω is $2^{10} = 1024$.

- The event A contains all the element $(c_1, c_2, \ldots, c_{10})$ of Ω such that at least one c_i is equal to H.

  $$A = \left\{(c_1, c_2, \ldots, c_{10}) \mid c_i = H \text{ and for } j \neq i, c_j = H \text{ or } T,\ j = 1, \ldots, 10, \right\}.$$

Events as sets

Solution(cont.)

Some examples of outcomes from A : (T,T,H,T,H,H,T,T,T,T), (T,T,H,H,H,H,T,T,H,T), (T,T,H,T,H,H,T,T,H,H).
Remark that only one element of Ω is not in A; that is (T,T,T,T,T,T,T,T,T,T).

- The event T_j contains all the element $(c_1,c_2,\ldots,c_{10})$ of Ω such that $c_j = T$.

$$T_j = \{(c_1,\ldots,c_{j-1},T,c_{j+1},\ldots,c_{10}) \mid \text{for } i \neq j, c_i = H \text{ or } T, i = 1,\ldots,10.\}.$$

Some examples of outcomes from T_j :
$\omega_1 = (T,T,H,T,T,H,T,T,T,T) \in T_5$,
$\omega_2 = (T,T,H,H,H,H,T,T,H,T) \in T_2$,
$\omega_3 = (T,T,H,T,H,H,T,T,H,H) \in T_7$.
Remark: ω_1, ω_2 and ω_3 belong to T_8.

Examples of random experiment (cont.)

- $\{$*Measuring some physical quantity*$\}$, for example, temperature, is a random experiment.
- In this case, Ω is $\mathbb{R} =]-\infty, +\infty[$, even if in practise temperature has a lower bound and an upper bound.
- Events are ranges of temperature as intervals in $\mathbb{R}$.
- For example, the event A that the temperature is larger than 12 but less than or equal to 23 is $A =]10, 23]$. The event that the temperature is below 0 is $]-\infty, 0[$.

Events as sets

Let A and B be two subsets of Ω.
The event *'A **or** B occur'* is the subset $A \cup B$.

Definition

Union of two sets A and B:

$$A \cup B = \{\omega \in \Omega \mid \omega \in A \text{ or } \omega \in B\}.$$

The event *'A **and** B occur'* is the subset $A \cap B$.

Definition

Intersection of two sets A and B :

$$A \cap B = \{\omega \in \Omega \mid \omega \in A \text{ and } \omega \in B\}.$$

Events as sets

Example:
Drawing a card from a deck, considering the events
A: *'the card is the king of diamonds'*
B: *'the card is the ace of hearts'*

- obtaining A and B simultaneously is **impossible!**

An impossible event is represented by the empty set.

Definition

The empty set, denoted $\emptyset$, is the unique set such that for all $\omega \in \Omega$, $\omega \notin \emptyset$.

In other words, $\emptyset$ has no element.
Remark: $A \cap \emptyset = \emptyset$, $\forall A \subseteq \Omega$.

Events as sets

The event *'A does not occur'* means that the outcome belongs to the complement of A.

Definition

The complement of a set A is

$$\overline{A} = \{\omega \in \Omega \mid \omega \notin A\}.$$

Remarks:

- $A \cap \overline{A} = \emptyset$ and $A \cup \overline{A} = \Omega$.
- Performing an experiment leads necessarily to an outcome which is in Ω
  - ➥ Ω is the certain event.

Events as sets

The event *'A occurs, but not B'* is the set difference between A and B.

Definition

Difference between two sets A and B (set difference):

$$A \setminus B = A \cap \overline{B}.$$

Exercise: *'A and B do not occur simultaneously'*. Use the set theory jargon to describe this event?

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'A and B cannot occur simultaneously' means that A and B are disjoint.

Definition

A and B are disjoint iff $A \cap B = \emptyset$.

Events as sets

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Events as sets

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- $\overline{\overline{A}} = \ A.$
- $B \subseteq A \Longleftrightarrow A \cap B = \ B$
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- De Morgan's Laws : $\overline{A \cup B} =$

Events as sets

- $A \cup \emptyset = A$ ➥ $\emptyset$ acts as an **identity element** for the union.
- $A \cap \Omega = A$ ➥ Ω acts as an **identity element** for the intersection.
- $\overline{\overline{A}} = A$.
- $B \subseteq A \Longleftrightarrow A \cap B = B$
  $B \subseteq A \Longleftrightarrow A \cup B = A$.
- De Morgan's Laws : $\overline{A \cup B} = \overline{A} \cap \overline{B}$ and $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

  Proof?

Events as sets

Proof of De Morgan's laws:

- For any ω in Ω, if $\omega \in \overline{A \cup B}$, $\omega \notin A \cup B$ which means that ω is neither in A nor in B. Therefore $\omega \in \overline{A}$ **and** $\omega \in \overline{B}$. Thus $\omega \in \overline{A} \cap \overline{B}$. Then $\overline{A \cup B} \subseteq \overline{A} \cap \overline{B}$.
- Conversely, if $\omega \in \overline{A} \cap \overline{B}$, $\omega \notin A$ **and** $\omega \notin B$. Then $\omega \notin A \cup B$. Therefore $\omega \in \overline{A \cup B}$. Thus $\overline{A} \cap \overline{B} \subseteq \overline{A \cup B}$.

Conclusion:

$$\overline{A \cup B} = \overline{A} \cap \overline{B}.$$

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Thus $\omega \in \overline{A} \cap \overline{B}$. Then $\overline{A \cup B} \subseteq \overline{A} \cap \overline{B}$.

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Therefore $\omega \in \overline{A \cup B}$. Thus $\overline{A} \cap \overline{B} \subseteq \overline{A \cup B}$.

Conclusion:

$$\overline{A \cup B} = \overline{A} \cap \overline{B}.$$

Applying this result to $\overline{A}$ and $\overline{B}$, we have:

$$\overline{\overline{A} \cup \overline{B}} = \overline{\overline{A}} \cap \overline{\overline{B}} \Longleftrightarrow \overline{\overline{A} \cup \overline{B}} = A \cap B.$$

Taking the complement of each side of the last equality we obtain:

$$\overline{A} \cup \overline{B} = \overline{A \cap B}.$$

Events as Sets

Events	Set
Universe (sample space)	Ω
Outcome (elementary event)	$\{\omega\}$
Impossible event (null event)	$\emptyset$
Both A and B occur	$A \cap B$
Either A or B or both occur	$A \cup B$
A occurs but not B	$A \setminus B$
A and B are incompatible	$A \cap B = \emptyset$
A produces B ($A \Longrightarrow B$)	$A \subset B$

Events as sets

Exercise

Let A, B, C be 3 events.

Consider $E_1 = A \cap \overline{B} \cap \overline{C}$ and $E_2 = A \cap (B \cup C)$.

1. Show that E_1 and E_2 are disjoint.
2. What is $E_1 \cup E_2$?

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Solution

1. E_1 and E_2 are disjoint is equivalent to $E_1 \cap E_2 = \emptyset$

$$\begin{aligned} E_1 \cap E_2 &= (A \cap \overline{B} \cap \overline{C}) \cap (A \cap (B \cup C)) = A \cap \left[(\overline{B} \cap \overline{C}) \cap (B \cup C)\right] \\ &= A \cap \left[\overline{B \cup C} \cap (B \cup C)\right] = A \cap \emptyset = \emptyset. \end{aligned}$$

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2.

$$\begin{aligned} E_1 \cup E_2 &= (A \cap \overline{B} \cap \overline{C}) \cup (A \cap (B \cup C)) = A \cap \left[(\overline{B} \cap \overline{C}) \cup (B \cup C)\right] \\ &= A \cap \left[(\overline{B \cup C}) \cup (B \cup C)\right] = A \cap \Omega = A. \end{aligned}$$

σ-field

Events are subsets of Ω but in some cases, all subsets of Ω are not events.
We will consider a subcollection $\mathscr{F}$ of the set of all subsets of Ω.

Expected properties for $\mathscr{F}$

- if A, $B \in \mathscr{F}$ then $A \cup B \in \mathscr{F}$,
- if $A \in \mathscr{F}$ then $\overline{A} \in \mathscr{F}$,
- the empty set $\emptyset$ belongs to $\mathscr{F}$.

Any collection $\mathscr{F}$ of subsets of Ω which satisfies these three conditions is called a ***field*** or an ***algebra***.

Consequently ➥ If $A_1, A_2, \ldots, A_n \in \mathscr{F}$ then $\displaystyle\bigcup_{i=1}^{n} A_i \in \mathscr{F}$.

σ-field

In order to be able to discuss probability we need more; we require that the collection of events be closed under the operation of taking any countable unions of elements from $\mathscr{F}$.

➥ Then the collection becames a σ-field.

Definition

*A collection $\mathscr{F}$ of subsets of Ω is called a **σ-field** if it satisfies the following conditions:*

- $\emptyset \in \mathscr{F}$,
- *if* $A_1, A_2, \ldots \in \mathscr{F}$ *then* $\bigcup_{i=1}^{\infty} A_i \in \mathscr{F}$,
- *if* $A \in \mathscr{F}$ *then* $\overline{A} \in \mathscr{F}$.

σ-field

Exercise 1

Let A be a subset of Ω.

Prove that $\mathscr{F} = \{\emptyset, A, \overline{A}, \Omega\}$ is a σ-field.

σ-field

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Let A be a subset of Ω.
Prove that $\mathscr{F} = \{\emptyset, A, \overline{A}, \Omega\}$ is a σ-field.

Solution
The conditions are easily checked

- The empty set is in $\mathscr{F}$.
- We have for unions:

$$\begin{aligned}
&\emptyset \cup A = A \in \mathscr{F}, \quad A \cup \overline{A} = \Omega \in \mathscr{F}, \quad \emptyset \cup A \cup \overline{A} = \Omega \in \mathscr{F},\\
&\emptyset \cup \overline{A} = \overline{A} \in \mathscr{F}, \quad A \cup \Omega = \Omega \in \mathscr{F}, \quad \emptyset \cup A \cup \Omega = \Omega \in \mathscr{F},\\
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&A \cup \overline{A} \cup \Omega = \Omega \in \mathscr{F}, \quad \emptyset \cup A \cup \overline{A} \cup \Omega = \Omega \in \mathscr{F}.
\end{aligned}$$

- Complement:

$$\overline{\emptyset} = \Omega \in \mathscr{F}, \quad \overline{A} \in \mathscr{F}, \quad \overline{\overline{A}} = A \in \mathscr{F}, \quad \overline{\Omega} = \emptyset \in \mathscr{F}.$$

σ-field

Exercise 2

Let us consider $\Omega = \{a, b, c\}$.

1. Find $\mathscr{F}_s$, the smallest σ–algebra on Ω.
2. Find $\mathscr{F}_\ell$, the largest σ–algebra on Ω.
3. How many elements does the largest σ–algebra contain?

σ-field

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2. Find $\mathscr{F}_\ell$, the largest σ–algebra on Ω.
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Solution

1. $\mathscr{F}_s = \{\emptyset, \Omega\}$.
2. $\mathscr{F}_\ell$ contains necessarily $\emptyset$. Then it contains Ω.
   Assume $\{a\} \in \mathscr{F}_\ell$ then $\overline{\{a\}} = \{b, c\}$ is in $\mathscr{F}_\ell$.
   In the same way $\{b\}$, $\{a, c\}$ and $\{c\}$, $\{a, b\}$ belong to $\mathscr{F}_\ell$.
   Therefore $\mathscr{F}_\ell = \Big\{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}\Big\}$.
3. $\mathscr{F}_\ell$ contains 8 elements